

SERVER SCRIPT

CREATE_SMART_BOM

Script Type: API
Event Frequency: All
DocType Event: Before Insert
API Method: create_smart_bom

Script

```
# SCRIPT: SMART BOM CREATOR (FINAL WORKING VERSION)
# TYPE: API
# METHOD NAME: create_smart_bom
```

1. GET INPUTS

```
req_item_code = frappe.form_dict.get('item_code')
req_quality = frappe.form_dict.get('quality')
req_mb_item = frappe.form_dict.get('mb_item')
req_ldr = frappe.form_dict.get('ldr')
```

```
if not req_item_code or not req_ldr:
```

```
    frappe.response['message'] = "Error: Missing Input (Item or LDR)"
```

```
else:
```

```
    ldr_percent = float(req_ldr)
```

2. RAW MATERIALS (Item Codes)

```
ITEM_PP = "PP - 1002001"
```

```
ITEM_FL = "FL - 1003011"
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ITEM_PPA = "SA - 1004001"
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```
ITEM_ANTI = "SA - 1004002"
```

3. DEFINE MIXING RATIOS

```
mix_map = {
    "DELUXE": [35, 100, 1.0, 0.5],
    "PREMIUM": [100, 0, 0.5, 0.2],
    "GOLD": [100, 25, 0.5, 0.2],
    "BRONZE": [100, 35, 0.2, 0.2],
    "CLASSIC": [80, 40, 0.5, 0.2],
    "SUPER ECO": [100, 30, 0.5, 0.2],
    "PLATINUM": [100, 10, 0.5, 0.2],
    "ECO SPECIAL": [55, 60, 0.5, 0.2],
    "ECO GREEN": [60, 75, 1.0, 0.5],
    "LIFESTYLE": [65, 50, 0.5, 0.2],
    "SILVER": [100, 30, 0.5, 0.2],
    "SUPER PLATINUM": [100, 15, 0.5, 0.2],
    "SUPER CLASSIC": [75, 50, 0.5, 0.2],
    "ULTRA": [40, 75, 1.0, 0.5]
```

```
}
```

```

q_key = str(req_quality).upper().strip()
ratios = mix_map.get(q_key, [100, 35, 0.2, 0.2])

parts_pp = ratios[0]
parts_fl = ratios[1]
parts_ppa = ratios[2]
parts_anti = ratios[3]

# 4. CALCULATE WEIGHTS (3 Decimal Precision)
base_tot = parts_pp + parts_fl
mb_weight = (base_tot * ldr_percent) / 100.0
total_batch_weight = base_tot + parts_ppa + parts_anti + mb_weight
one_mixing_factor = 1.0 / total_batch_weight

t_pp = round(parts_pp * one_mixing_factor, 3)
t_fl = round(parts_fl * one_mixing_factor, 3)
t_ppa = round(parts_ppa * one_mixing_factor, 3)
t_anti = round(parts_anti * one_mixing_factor, 3)
t_mb = round(mb_weight * one_mixing_factor, 3)

# 5. SMART CHECK (Including Custom LDR Field)
existing_boms = frappe.get_all("BOM",
    filters={
        "item": req_item_code,
        "is_active": 1,
        "docstatus": 1,
        "custom_ldr_": ldr_percent # Matching based on your custom field
    },
    fields=["name"]
)

match_found = None

for b in existing_boms:
    items = frappe.get_all("BOM Item", filters={"parent": b.name}, fields=["item_code", "qty"])

    # Check specific quantities to identify the recipe
    check_pp = [x for x in items if x.item_code == ITEM_PP and abs(x.qty - t_pp) < 0.001]
    check_fl = [x for x in items if x.item_code == ITEM_FL and abs(x.qty - t_fl) < 0.001]
    check_ppa = [x for x in items if x.item_code == ITEM_PPA and abs(x.qty - t_ppa) < 0.001]
    check_anti = [x for x in items if x.item_code == ITEM_ANTI and abs(x.qty - t_anti) < 0.001]
    check_mb = [x for x in items if x.item_code == req_mb_item and abs(x.qty - t_mb) < 0.001]

    found_pp = check_pp[0] if check_pp else None
    found_fl = check_fl[0] if check_fl else None
    found_ppa = check_ppa[0] if check_ppa else None

```

```

found_ppa = check_ppa[0] if check_ppa else None
found_anti = check_anti[0] if check_anti else None
found_mb = check_mb[0] if check_mb else None

if found_pp and found_ppa and found_anti and found_mb:
    if t_fl > 0 and found_fl:
        match_found = b.name
        break
    elif t_fl == 0 and not found_fl:
        match_found = b.name
        break

if match_found:
    frappe.response['message'] = match_found
else:
    # 6. CREATE NEW BOM
    bom = frappe.new_doc("BOM")
    bom.item = req_item_code
    bom.quantity = 1.0
    bom.is_default = 1
    bom.is_active = 1
    bom.currency = "INR"
    bom.rm_cost_as_per = "Valuation Rate"

    # SET CUSTOM FIELD VALUE
    bom.custom_ldr_ = ldr_percent

    bom.append("items", {"item_code": ITEM_PP, "qty": t_pp, "uom": "Kg"})
    if t_fl > 0:
        bom.append("items", {"item_code": ITEM_FL, "qty": t_fl, "uom": "Kg"})
    bom.append("items", {"item_code": ITEM_PPA, "qty": t_ppa, "uom": "Kg"})
    bom.append("items", {"item_code": ITEM_ANTI, "qty": t_anti, "uom": "Kg"})
    bom.append("items", {"item_code": req_mb_item, "qty": t_mb, "uom": "Kg"})

    bom.insert(ignore_permissions=True)
    bom.submit()
    frappe.response['message'] = bom.name

```

Request Limit:

5

Time Window (Seconds):

86400